Assignment - 1
(Submit on 10/March/2026 during the class)

1. Solve the X-OR gate problem with a 1-hidden layer MLP with the hidden layer containing 2 neurons. Use the Leaky-ReLU activation function in the hidden layer and the sigmoid activation function in the output layer with Binary Cross Entropy as the loss function.
    a. Derive the formula for the weight update for both layers using stochastic gradient descent optimization
    b. Use the random initialization for weights and biases
    c. Write the updated weights/biases and loss after two epochs
2. What is the weight updation rule for a perceptron model with **sign** activation function? Solve the OR gate problem for two epochs with this weight updation rule.
3. Create an MLP with multiple hidden layers and calculate the number of learnable parameters. (Try for at least 3 different combinations of hidden layers and 3 different combinations of the neurons within.)
4. Discuss the role of learning rate in training MLPs. What are the consequences of setting the learning rate too high or too low? What adaptive learning rate strategies can be used for better learning?
5. Consider a 5-layer MLP trained using sigmoid activation and standard backpropagation. Show mathematically why deeper neural networks with sigmoid activation suffer from vanishing gradient problems. What will happen if the sigmoid is replaced with ReLU activations?
6. Training an MLP with 100 million parameters on a large dataset is computationally intensive. If memory and computation were limited, how would you design an approximate learning algorithm that reduces complexity without major loss in accuracy?
7. Standard activation functions like sigmoid, ReLU, tanh, etc. have their own limitations. Design a new activation function that combines the best properties of existing ones. Derive its mathematical formulations and prove whether it satisfies the key properties (differentiability, boundedness, etc.)
8. Supervised learning requires labelled data, which is expensive. Can an MLP be trained without labelled data (unsupervised/self-supervised learning) and still perform classification/regression?
9. Some researchers argue that backpropagation is biologically implausible for deep learning. Critically evaluate this claim and propose an alternative non-backpropagation-based training method that could work well for MLPs.
10. Derive the formulae for He and Xavier's weight initialization methods.
11. State and prove the Universal Approximation Theorem for the one-dimensional case.

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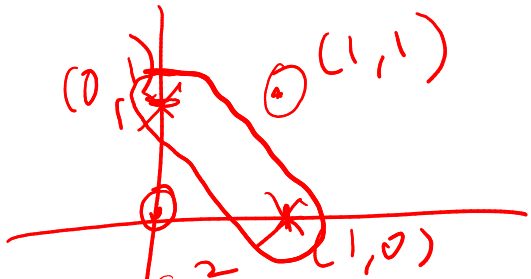
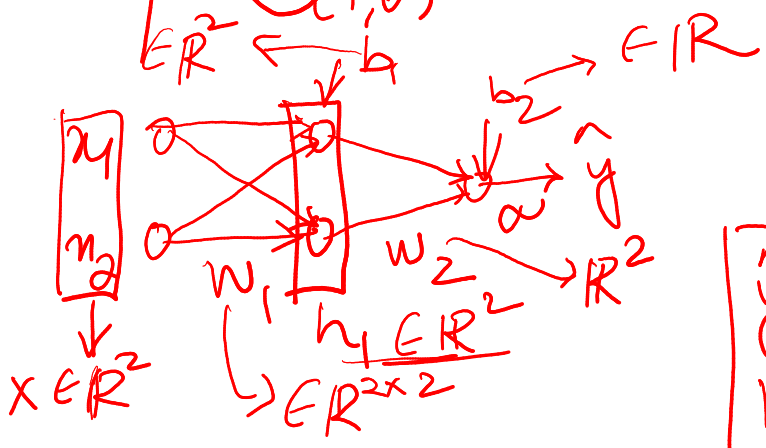


x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0



$$\boxed{\begin{aligned} \hat{y} &= \sigma(w_2^T h_1 + b_2) \\ h_1 &= LR(w_1^T x + b_1) \end{aligned}}$$

$$\theta_{t+1} = \theta_t - \eta \nabla_\theta l(\theta)$$

$$\theta = \{w_1, w_2, b_1, b_2\}$$

$$l(\theta) = -\left[y \log(\hat{y}) + (1-y)\log(1-\hat{y})\right]$$

$$\nabla_\theta l(\theta) = \frac{\partial l}{\partial w_2} = \frac{\partial l}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial w_2} = (\hat{y} - y) h_1$$

$$(w_2)_{t+1} = (w_2)_t - \eta (\hat{y} - y) h_1$$

$$(b_2)_{t+1} = (b_2)_t - \eta (\hat{y} - y)$$

$h_1 = LR(z) = f(z)$

$z = w_1^T x + b_1$

$$\underset{2\times 2}{(w_1)_{t+1}} = \underset{2\times 2}{(w_1)_t} - \eta \underset{1\times 1}{\times} \underset{2\times 1}{\left[\left\{\underset{1\times 1}{(\hat{y} - y)}\, \underset{2\times 1}{w_2}\right\} \odot \underset{(2\times 1)}{f'(z)}\right]^T}$$

$\times$ matrix

$(2\times 1) \odot (2\times 1)$

$(2\times 1)^T$

(1×2)

MM

2×2

$$(b_1)_{t+1} = (b_1)_t - \eta \left[\left\{(\hat{y} - y) w_2\right\} \odot f'(z)\right]$$